

MeshEcho-CPR: Cost-Responsive Recovery with ACK-Terminated Relay Suppression for LoRa Mesh Networks

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Abstract—Packet loss in a LoRa mesh creates a choice between repeating a known path and injecting another network-wide flood. Repeating too early wastes a stale route, whereas flooding too freely multiplies airtime and collisions. We present MeshEcho-CPR, a source and relay policy that makes this choice at physical transmission time. A routed packet is retried once on the same confirmed path; only after that retry has physically started can one deadline-feasible recovery flood be admitted. A relay records a pending flood and cancels it only after a physically decoded ACK matches the flow, request, endpoints, reverse path, repair marker, and sender, and only before the relay commits the transmission. This preserves a charged wire cost while suppressing redundant relay floods. In an audited 20-seed holdout with 20 nodes and 6-dB block fading, CPR reaches 0.9829 deadline ACK PDR and 0.9984 destination PDR with 6.29 s mean TX airtime. Against a no-rescue control, it improves ACK PDR by 0.0334 [0.0172, 0.0495] and destination PDR by 0.0364 [0.0191, 0.0538] while reducing airtime by 3.14 s [2.53, 3.76]. Against an identical no-cancel ablation, its reliability is statistically comparable but airtime is 4.19 s lower. Results are packet-level simulation evidence for this workload, not measurements of deployed Meshtastic or MeshCore hardware.

Index Terms—LoRa mesh, opportunistic recovery, ACK feedback, flooding suppression, deadline-aware routing, packet-level simulation

I. INTRODUCTION

LoRa is attractive for low-power and infrastructure-sparse deployments because it provides a long link budget with inexpensive radios, but its low data rate makes redundant transmissions costly [1]. Multihop LoRa systems therefore combine cached routes, flooding, and local forwarding decisions [2]–[4]. The hard case is a route that was recently confirmed but is no longer reliable: an immediate flood can recover delivery, yet every relay that transmits the same recovery packet consumes airtime and increases collision exposure.

Existing route metrics such as ETX quantify expected transmissions [5], but a metric alone does not specify when a recovery is safe to start or how a relay should react to a terminal ACK that is already on the air. This paper asks whether a protocol can use the physical ordering of repeat, recovery, ACK, and relay-commit events to improve reliability without paying for an unrestricted second flood.

We make three contributions:

- We define MeshEcho-CPR, a bounded source/recovery state machine that separates an initial confirmed-route attempt, one same-path repeat, and at most one deadline-gated recovery flood.

- We add an ACK-terminated relay ledger. Cancellation is accepted only for a physically matching ACK and an uncommitted pending relay, making the suppression event auditable rather than an oracle-based delivery shortcut.
- We provide paired development and untouched holdout evidence with a no-cancel ablation and two DRC controls. All arms share topology, traffic, radio, request denominators, and per-seed identities; the paper reports seed-level confidence intervals and complete TX cost.

The contribution is a rewritten source, recovery, and feedback core on a preserved packet-level LoRa substrate. The substrate supplies propagation, collision, forwarding, ACK transport, ToA, energy accounting, and evidence ledgers so that the protocols are compared under the same physical model. We do not claim a firmware implementation or universal superiority over every mesh protocol.

II. RELATED WORK

Simulation and measurement studies have established LoRa link, coverage, and collision models [6]–[8], while recent work has examined multihop routing and practical mesh libraries [9]–[11]. These systems expose a recurring trade-off: route state reduces redundant transmissions when it is valid, while flooding tolerates a missing or stale route at a higher channel cost. CPR focuses on the transition between these states rather than proposing another static path score.

The closest conceptual baselines in this study are a deadline-reserved source controller and its no-rescue form. They use the same packet model and ACK transport but do not use CPR’s repeat-before-recovery ordering or its ACK-terminated relay ledger. Thus the ablation isolates relay cancellation, whereas the no-rescue control isolates the value of allowing the recovery branch. The controls are algorithms in the simulator, not claims about any named product or firmware.

III. MESH ECHO-CPR

A. Local state and action ordering

Each unicast flow has a source, destination, enqueue time, original deadline, and a bounded route record. A route is usable only when its endpoint sequence, generation, expiry, and local state are valid. At the actual queue-available TX start, the source computes conservative upper bounds B_R and B_F for a routed DATA/ACK exchange and a recovery FLOOD/ACK exchange, including packet ToA, forwarding

delay, and a safety margin. The source does not inspect a future reception outcome or a global PRR oracle.

The initial action is R_{initial} when a confirmed route fits before the original deadline; otherwise it is F_{initial} when a flood fits. After an unacknowledged routed initial transmission, CPR schedules one same-path repeat. The repeat request is not considered a transmission until the packet physically starts. Only then can the recovery decision admit one flood, and only if the complete bound still fits:

$$\begin{aligned} R_{\text{initial}} &\rightarrow R_{\text{repeat}}, \\ R_{\text{repeat}} &\rightarrow F_{\text{recovery}} \quad \text{iff } t + B_F \leq t_{\text{deadline}}. \end{aligned} \quad (1)$$

If the flow is already acknowledged, the route generation is stale, or the bound does not fit, no recovery request is started. This ordering prevents a queued but never-started repeat from authorizing a flood.

B. ACK identity and route generation

An ACK is accepted at the source only when its request and flow identifiers, endpoints, reverse path, path index, physical sender, deadline, and repair marker agree with the flow record. The unmarked ACK corresponds to the initial attempt, marker 1 to the repeat, and marker 2 to the recovery flood. The ACK must refer to a packet that physically started; its RX attempt and ACK TX are recorded in the evidence ledger. A valid ACK commits the route generation and retires the flow. An ACK from a previous generation cannot mutate a newer route.

C. ACK-terminated relay suppression

When a relay decodes a CPR flood, it creates a bounded pending handle indexed locally by relay, origin, flow, request, and marker. The handle stores the endpoints and exact path alongside the queued FLOOD request. A relay cancels that request only when

$$\begin{aligned} \text{cancel} &= \text{match}(\text{flow}, \text{request}, \text{endpoints}) \\ &\quad \wedge \text{match}(\text{path}, \text{marker}, \text{sender}) \wedge \neg \text{committed}. \end{aligned} \quad (2)$$

The path must be loop-free and the decoded ACK must be on the corresponding reverse hop. A wrong flow, stale generation, malformed path, application payload, or already-committed handle is rejected. The ledger is capped per node; capacity eviction is charged and logged. This makes every saved relay transmission attributable to a physical ACK event.

D. Controls

The primary ablation, CPR-no-cancel, runs the same source state machine and wire markers but never cancels a pending relay. DRC-no-rescue uses the same deadline-reserved route state without a recovery flood. DRC is the sealed rescue control from the independent development audit. All four arms use the same topology, application trace, channel draws, packet accounting, and scheduled-flow denominator for a given seed.

TABLE I
FROZEN CPR EXPERIMENT PARAMETERS.

Parameter	Value
Nodes / area	20 / $3000 \times 3000 \text{ m}^2$
Traffic	8 unicast flows/min, 4 cyclic pairs
Generation / drain	240 s / 270 s
Radio	SF7, 125 kHz, CR 4/5, 17 dBm
Payload / deadline	32 B / 30 s
Fading	4 dB spatial, 6 dB every 60 s
Max hops / route TTL	6 / 600 s

IV. EXPERIMENTAL METHOD

A. Frozen case and cohorts

The case uses 20 nodes uniformly placed in a 3000 m square, four cyclic directed pairs, and 8 Poisson unicast flows/min for a 240-s generation horizon followed by a 30-s drain. The channel uses SF7, 32-byte payloads, 17-dBm transmit power, 125-kHz bandwidth, coding rate 4/5, 4-dB spatial shadowing, and 6-dB temporal block fading refreshed every 60 s. The maximum path length is six hops, route TTL is 600 s, and application deadline is 30 s. Table I summarizes the fixed inputs.

The development cohort contains seeds 92200–92219 and was used to freeze the analysis gates. The untouched holdout contains seeds 92300–92319 and was opened only after the development manifest and gate hash passed. Every arm has the same scheduled unicast denominator across each cohort (676 aggregate flows across the 20 development seeds, 668 aggregate flows across the 20 holdout seeds; these are cohort totals, not per-seed counts). No smoke or exploratory seed is pooled into the reported estimates.

B. Metrics and audit

The primary reliability metrics are deadline ACK PDR at the source and deadline destination PDR. Cost is complete network TX airtime and modeled TX energy for packets physically started before the drain cutoff; relay and ACK packets are included. We report the arithmetic mean of the 20 seed-level values. For each comparator, per-seed CPR-minus-control differences are paired by seed and summarized with a nominal two-sided 95% Student- t interval (19 degrees of freedom). Flows, requests, physical TXs, per-receiver RX attempts, ACK provenance, repair markers, and route generations are joined by immutable identifiers. An incomplete join invalidates the cohort.

V. RESULTS

A. Holdout performance

Table II gives the untouched holdout means. CPR has the highest ACK PDR among the four arms within rounding and nearly saturated destination PDR, while using less TX airtime than both rescue controls. The no-cancel arm has almost the same reliability but 66.7% more airtime than CPR, showing that the relay ledger is the source of the cost reduction rather than a different traffic trace.

TABLE II
HOLDOUT MEANS OVER 20 PAIRED SEEDS. PDRs ARE DEADLINE-BASED; ENERGY IS THE SIMULATOR’S MODELED TX ENERGY, NOT A BATTERY MEASUREMENT.

Arm	ACK PDR	Dest. PDR	TX airtime (s)	TX energy (J)	TX count	RX attempts
CPR	0.9829	0.9984	6.290	2.491	81.65	1633
CPR-no-cancel	0.9734	0.9985	10.485	4.152	124.30	2486
DRC-no-rescue	0.9495	0.9619	9.433	3.736	112.45	2249
DRC	0.9810	0.9985	10.859	4.300	128.20	2564

TABLE III
PAIRED HOLDOUT DIFFERENCES, CPR MINUS COMPARATOR. ENTRIES ARE MEAN DIFFERENCE [NOMINAL TWO-SIDED 95% CI] OVER 20 SEEDS.

Comparator	ACK PDR	Dest. PDR	TX airtime (s)	TX energy (J)
CPR-no-cancel	+0.0095 [−0.0028, +0.0218]	−0.0001 [−0.0048, +0.0047]	−4.195 [−4.686, −3.704]	−1.661 [−1.856, −1.467]
DRC-no-rescue	+0.0334 [+0.0172, +0.0495]	+0.0364 [+0.0191, +0.0538]	−3.143 [−3.757, −2.529]	−1.245 [−1.488, −1.001]
DRC	+0.0018 [−0.0049, +0.0086]	−0.0001 [−0.0048, +0.0047]	−4.568 [−5.240, −3.897]	−1.809 [−2.075, −1.543]

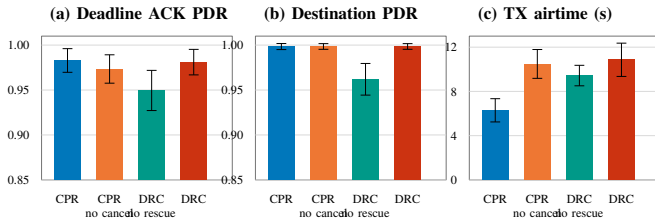


Fig. 1. Holdout means with nominal 95% seed-level confidence intervals. CPR maintains reliability while reducing TX airtime relative to both no-cancel and DRC controls.

B. Mechanism and development gates

The holdout CPR arm physically started 584 initial routed attempts, 26 repeats, and 21 recovery floods; it also recorded 1,606 ACK-terminated relay cancellations. These counts demonstrate that the recovery branch and the ledger were exercised rather than remaining dormant. The development cohort started 589 initial routed attempts, 40 repeats, and 26 recovery floods, above the preregistered exposure minima. Its reliability lower bounds were not below -0.05 against either control, and mean CPR airtime was 28.2% below DRC-no-rescue and below no-cancel. The holdout preserved the same direction: CPR was 34.2% below DRC-no-rescue and 40.0% below no-cancel in mean TX airtime.

The no-cancel comparison is especially diagnostic. Its holdout ACK PDR is 0.9734 versus CPR’s 0.9829, with a paired interval containing zero, but it starts 42.7 more TX packets per run and receives 853 more RX attempts on average. The result supports a cost effect from terminating uncommitted relay work, not a claim that cancellation alone guarantees higher reliability.

C. Physical cost and state exposure

The holdout cost difference is visible in the physical ledgers, not only in a derived PDR. CPR averages 81.65 started TX packets, 1,633 receiver attempts, and 119.75 collision failures per seed. CPR-no-cancel averages 124.30 TX packets, 2,486 receiver attempts, and 654.55 collision failures. DRC-

no-rescue averages 112.45 TX packets and 566.30 collision failures, while DRC averages 128.20 TX packets and 707.60 collision failures. The lower collision count for CPR is consistent with fewer pending relay floods, but the experiment does not claim that cancellation is the sole cause of every collision difference: changing a packet schedule changes later interference as well.

The operational state is finite. In the evaluated configuration each source keeps at most eight route records and eight active-flow records, and each relay keeps at most eight cancelable CPR flood handles. The broader simulator also caps pending relay fallback handles and records capacity evictions. A handle is retired after commit, cancellation, ACK acceptance, or deadline censoring; there is no unbounded protocol cache hidden from the audit. These bounds make the algorithm implementable on constrained nodes and make state drops visible in the request ledger rather than silently changing the denominator.

D. Event ordering and causal boundary

The evidence contract gives equal-time events an explicit order: a physical TX end and its receiver attempts are processed before ACK decisions, deadline censoring, application enqueue, guard/recovery timers, and a new TX request. Consequently an ACK can cancel a pending relay only before that relay’s request commits. A packet that has already started remains fully charged even if its ACK arrives later or after the application deadline. This boundary is important for the causal interpretation: CPR observes local physical ACK evidence and queue state, but never reads the simulator’s final destination delivery flag to decide whether to cancel.

The source decision and relay ledger are both linear in the path length for a single packet. Route and flood bounds enumerate the modeled wire exchanges; ACK validation checks the encoded path and one reverse hop; relay cancellation uses a bounded keyed lookup. Thus the additional control logic does not add a network-wide coordination round or a second discovery protocol.

VI. DISCUSSION AND LIMITATIONS

CPR improves the reliability-cost trade-off relative to the no-rescue control in the declared fading workload and reduces redundant relay work relative to its matched no-cancel ablation. Its destination PDR is statistically indistinguishable from no-cancel and the full DRC control in the holdout, so the main evidence is cost reduction at comparable reliability rather than universal delivery superiority.

The study is a packet-level simulation. It models path loss, shadowing, half-duplex collisions, capture, LoRa ToA, forwarding delay, and ACK provenance, but not modem implementation details, antenna variation, duty cycle enforcement, waveform-level interference, or measured battery current. The energy values are modeled TX energy and exclude a device-specific battery experiment. The four directed pairs and 20-seed cohort represent one repeated traffic workload; mobility, larger networks, different spreading factors, and bursty or broadcast traffic may change the result. The DRC controls are algorithmic controls in the same harness, not implementations of a named standard or product. A hardware or hardware-in-the-loop validation is needed before deployment claims.

The ACK-terminated ledger also has a deliberate boundary: a relay can cancel only before its FLOOD request commits. It cannot undo an already-transmitted packet. This keeps the cost accounting conservative, but it means the benefit depends on the relative timing of ACK propagation and relay backoff. Future work should test that timing on measured channels and compare against a fixed-candidate route metric under a separately frozen workload.

A. Interpretation of the controls

The two DRC arms separate two questions that are otherwise easy to conflate. DRC-no-rescue keeps the DRC reserved-route state but removes its recovery flood. CPR also uses a different initial-action rule, so the positive CPR-minus-DRC-no-rescue reliability interval is evidence for the combined CPR source/recovery policy under this workload, not an isolated estimate of the recovery sequence alone. CPR and full DRC have nearly identical holdout PDRs, but CPR uses 4.57 s less TX airtime per seed. This is evidence for a more selective policy, not proof that every DRC decision is inferior: the controls still differ in source state, timer placement, and relay behavior.

The no-cancel arm is the stronger causal comparison for the relay ledger. It shares CPR’s initial action, repeat, recovery marker, deadline, and route generation rules. Its reliability interval contains zero while its airtime and energy intervals are strictly positive for CPR-minus-control in the negative direction. The result supports the narrower statement that matching ACK evidence can terminate pending relay work at lower cost under the frozen timing model. It does not establish that cancellation improves PDR in every topology, because a cancellation can also remove a transmission that would have succeeded if the ACK had arrived later.

These controls also give a testable transfer prediction. On a channel where ACKs arrive after most relay backoffs have

TABLE IV
AUDITED ARTIFACT ROLES USED BY THE PAPER.

Artifact	Required invariant
Manifest	Fixed arm order, seed split, case, source hashes, and paired trace identities.
Runs CSV	One row per seed and arm; equal scheduled-flow denominator and complete cost totals.
JSONL ledgers	Flow, action, request, TX, RX-attempt, and ACK joins with unique physical IDs.
Gate report	Development passes before holdout; all reliability, exposure, and cost checks are true.
Figure summary	Recomputed equal-seed means and paired t intervals from raw runs, not hand-entered values.

already committed, CPR should retain the reliability ordering but lose much of its airtime advantage. On a channel with fast reverse ACKs and many redundant flood relays, the ledger should save more TXs. A measured-channel experiment can therefore falsify the cost claim without changing the source state machine or the statistical denominator.

VII. REPRODUCIBILITY

The source implementation is identified by the repository commit and the CPR source-hash fields in each manifest. The experiment contract records the case, arm order, seed split, topology hash, radio hash, application trace hash, and paired denominators. The raw development and holdout bundles contain CSV run summaries plus JSONL flow, action, request, TX, RX-attempt, and ACK provenance records. The tables and Fig. 1 are generated directly from those bundles using the checked-in figure builder; no rounded value was entered as an independent measurement. The full evidence audit rejects a missing arm, duplicate physical identifier, incomplete RX/TX join, source-hash mismatch, or a holdout without a passed development prerequisite.

The resulting package is intentionally reproducible at the algorithm and packet-model level. It is not a claim that a reader can reproduce a radio waveform or a battery trace from the CSV files. The distinction matters when interpreting the reported energy and when deciding whether a future hardware study should preserve the same deadline and relay-commit definitions.

VIII. CONCLUSION

We presented MeshEcho-CPR, a rewritten source/recovery and feedback core for LoRa mesh simulation. Its strict repeat-before-recovery ordering and physically matched ACK-terminated relay ledger reduce redundant FLOOD work without relying on a future delivery oracle. In the untouched 20-seed holdout, CPR improves deadline ACK and destination PDR over the no-rescue control while using 34.2% less TX airtime, and it uses 40.0% less airtime than an identical no-cancel arm at comparable reliability. These findings are bounded by the stated packet-level model and workload; they motivate physical validation rather than justify firmware-level or universal protocol claims.

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